

Reflection Paper on the Stuxnet Virus Incident through CIA Triage

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INFT1102: Information Security

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September 12, 2025

Introduction

The documentary “Zero Day” is a movie about the Stuxnet virus and how it was created primarily by the United States to target the Natanz nuclear facility in Iran and their Siemens Programmable Logic Controllers. Their motive for targeting this nuclear facility involved preventing them from creating nuclear weapons to ensure the safety of citizens in many countries from harm. We can use the confidentiality, integrity, and availability (CIA) triage to identify ways the Stuxnet virus could have been better contained from the point of view of security engineers at the Natanz Iranian Nuclear Facility. The following details will provide analysis to how the Stuxnet virus could have been prevented or mitigated.

CIA Triad

Confidentiality

Confidentiality refers to the protection of information from unauthorized access and that the data of an organization is kept privately. Firstly, the facility should not have allowed the released photographs of within the facility. These photographs were taken during a tour for the Iranian president, and it leaked information of computer screens and the centrifuge models that the nuclear facility used. The photos helped creators of the Stuxnet virus to create code to specifically target those centrifuges and disrupt their functions.

Access control policies for digital media such as USB drives could have also been implemented to prevent the Stuxnet virus from infecting the facility systems. Blocking all unauthorized devices, disabling autorun and autoplay features in operating systems, and ensuring read-only access (Arrospide) are controls that could have helped Stuxnet from affecting their facility systems.

Lastly, the facility could have ensured network segmentation was used wherever applicable. For example, the control systems of the centrifuges should be isolated from the administrative networks and any other unrelated network. There is a possibility that a USB drive infected with the Stuxnet virus could have been initiated in a far section of the network and automatically spread to the centrifuge systems.

Integrity

Integrity refers to ensuring that data is not unintentionally changed or erased. Stuxnet successfully corrupted the data that the centrifuges were sending back to the system controls. The operator screens showed that the centrifuges were spinning at the correct speeds when they were being sped up and slowed down, causing them great failure.

Among other forms of defense, an intrusion detection system or real-time monitoring could have been used to monitor traffic. Human operators would be able to see data from the centrifuges to the operator screens and identify any suspicious activity that the Stuxnet virus was performing.

Additionally, an independent monitor that would have been a secondary, offline method to see the spin of the centrifuges removes the possibly of altered measurements from displaying (How nozomi networks could have mitigated the Stuxnet attack). The automated systems could have been consistently supplemented with manual checks given the workers provided to perform them.

Availability

Availability guarantees that authorized users can access systems and data as required. Stuxnet successfully prevented the availability of the centrifuges by causing them to fail. Some actions of containment would be to implement a patch management policy and a manual emergency shutdown procedure.

The patch management policy would deal with the zero-day vulnerabilities that the Stuxnet virus took advantage of by updating security of the systems (Insights, 2024). The emergency shutdown procedures would aim to manually stop the centrifuges from spinning without the event of an explosion, crash, or any physical damage.

Additional ways that the security engineers could have contained the virus would be to harden the systems and stress the air gap enforcement. System hardening would involve ensuring that the bare minimum of software and services would be met and that all supplementary or unnecessary software would be removed. This would also ensure that applications are to be inspected and approved before they can run. Stressing the air gap enforcement would mean thoroughly scanning external devices such as USB drives before introducing them to the facility systems.

Conclusion

From the view of a security engineer at the Natanz Iranian Nuclear Facility, containing the Stuxnet virus proved a challenge that required a multi-layered defense that would closely follow the CIA triage. By implementing strict network segmentation, access controls, manual consistent validation, intrusion detection, hardening systems, and stressing air gap controls, the facility could have significantly reduce the impact of the attack. With these security actions, the nuclear facility would grow and continue to have production without delay.

Arrospide, J. (n.d.). *USB device control: Controlling USB devices: USB-Lock-RP*. USB.

[https://www.usb-lock-rp.com/usb-device-control.html#:~:text=Enforced%20USB%20Blocking%3A%20Prevents%20unauthorized,identifier%20\(VID%2FPID\)](https://www.usb-lock-rp.com/usb-device-control.html#:~:text=Enforced%20USB%20Blocking%3A%20Prevents%20unauthorized,identifier%20(VID%2FPID)).

How nozomi networks could have mitigated the stuxnet attack. NEXTGEN 3.0 - Homepage. (2025, June 10). <https://nextgen.group/blog-post/stuxnet-attack#:~:text=It%20was%20designed%20to%20specifically,in%20the%20Windows%20operating%20system>.

Insights, D. (2024, December 10). *Benefits of maintaining security and confidentiality at work*. <https://www.dataguard.com/blog/benefits-of-maintaining-security-and-confidentiality-at-work/>

What is the CIA triad and why is it important?. Fortinet. (n.d.). <https://www.fortinet.com/resources/cyberglossary/cia-triad>